THREE NEW BOOKS ON STRAWBALE AND NATURAL BUILDING

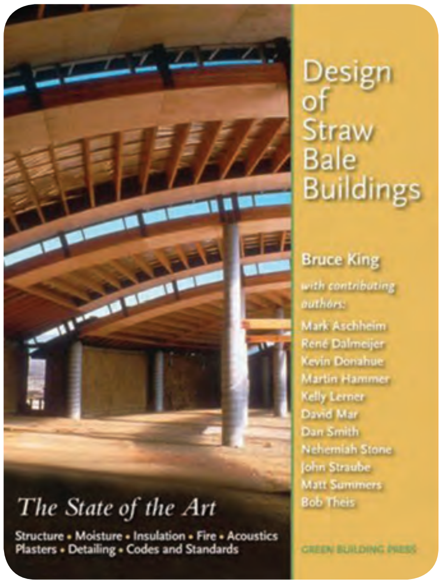
A new interest in energy efficiency helps account for the recent revival of natural building. A passive-solar building with earth-plastered straw-bale walls is simultaneously structure, insulation, and mass. The use of local, minimally processed materials and a sun-harvesting design reduces the carbon footprint of each new natural home constructed—especially its lifetime requirement for fossil fuel energy. And in this age of peak oil and global warming, energy-saving solutions top the list of design criteria for conscious architects and homeowners.

In the last decade, the mainstream world has awakened to the potential of strawbale construction, which has proven—through experimentation and testing—to be a viable insulating and structural wall system. And strawbale may be the cheapest, healthiest, most aesthetically versatile "new" material as well.

A variety of different post-and-beam and load-bearing systems have been used in construction, tested in laboratory settings, and adopted into building codes. Judging from the plethora of new books on green building, and specifically strawbale construction, the challenge now is to sort through the many structural approaches, plaster options, and aesthetic choices, not to mention sorting through the educational resources available. Three of the most recent books are described below.

Design of Straw Bale Buildings: The State of the Art

The latest book published has been literally years in the making. Hot off the press as of this writing, *Design of Straw Bale Buildings: The State of the Art* sums up the results of the many



laboratory and field tests conducted on strawbale wall systems worldwide to date. In chapters titled "Structure," "Moisture," "Fire," "Insulation," "Acoustics," "Plasters," "Detailing," and "Codes," author Bruce King thoroughly examines how strawbale stands up to contemporary standards for building materials. The book includes a section on best practices from a panel of expert contributors.

California engineer Bruce King has been at the forefront of the strawbale building movement for over a decade. The state's surplus of rice straw and its seismic potential create a unique regulatory climate that both encourages strawbale construction and enforces strict building codes. From the collective activity of California building professionals has come an impressive amount of strawbale test data, which King and his coauthors have updated, added to, and interpreted in this book.

Appropriate charts, graphs, and photographs are interspersed throughout the meaty text in a reader-friendly layout. King's writing style is intelligent, clear, and often witty. He injects levity into the technical discussions with quotes from the likes of Yogi Berra, and with an occasional cartoon.

The book describes the structural behavior of strawbale walls under compressive, lateral, shear, and seismic loads; with two- and three-tie bales, bales stacked flat, and bales stacked on edge. Reinforcing methods and plasters of all types are also described, with a refreshing emphasis on earthen and lime plasters, which King favors for their permeability and low embodied energy. (Cement contributes significantly to global warming by spewing a pound of CO_2 into the atmosphere for every pound of cement produced.)

For the serious seeker, King's discussion of clay, lime, gypsum, and cement plasters is the best I have read anywhere.

Equally impressive is the chapter on moisture, written by Canadian expert John Straube. Strawbale must be protected from water—in the form of precipitation, vapor, or ground moisture. Straube stresses "designing moisture problems out, not solving them after they have been needlessly designed into the enclosure." He discusses everything from the microscopic properties of water to site-specific approaches to "designing moisture out."

Some of these design strategies—such as sheltering roof overhangs and capillary breaks at the foundation—are well known. Straube also explains how water vapor moves through wall assemblies, and how excess moisture can be safely "stored" in natural plasters and mass walls, and later removed by evaporation. I wish all construction professionals could read and understand this information, which is critical to the longevity of all buildings.

Design of Straw Bale Buildings is an essential resource for engineers, architects, designers, and builders who need data to back up their design choices. But it is not just a reference for professionals. Not only does it explain the structural properties of strawbale, but it offers a fair-minded view of accepted practices and of the design choices available, based on the best information to date. Anyone thinking about building with straw should read this book.

King's book explains the theory of building with straw, and the best bale building practices. The next book offers a step-by-step how-to approach to the actual building process.

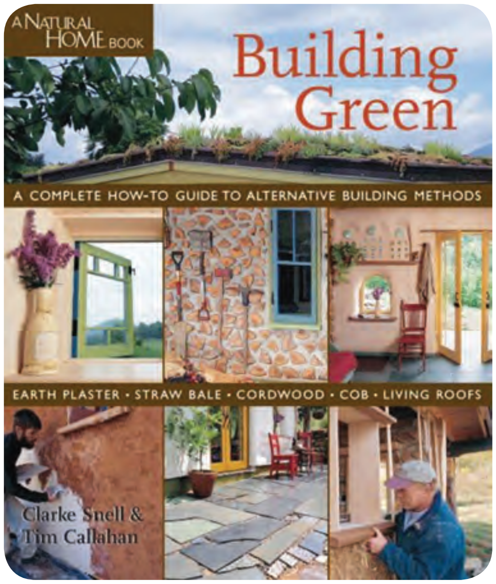
King, Bruce, *Design of Straw Bale Buildings: The State of the Art*. San Rafael, California: Green Building Press, 2006. 260 pp. paper. Illustrated. $40.

Building Green: A Complete How-to Guide to Alternative Building Methods

Building Green: A Complete How-to Guide to Alternative Building Methods, by Clarke Snell and Tim Callahan, chronicles the construction of a small guest house in North Carolina. Built as a practical demonstration, the house features four different wall systems: cob on the south, cordwood on the east, stick frame on the west, and strawbale on the north. Topped with a living roof, the charming cottage is a personalized work of art.

At first glance, with its comparison of different wall systems in the same building, *Building Green* seems like the how-to book we've all been waiting for. The layout is inviting, and the hundreds of full-color photographs offer clear, step-by-step visuals. The building itself incorporates many good ideas, and Snell amply explains the decision-making thought process.

Perhaps too amply. The book is over 600 pages and weighs over 4 lb; for me, just holding it up to read became a problem. The weight of the



information was also hard to process. Green building philosophy, structural basics, and design theory precede construction, and not until page 118 do they begin preparing the site.

Snell does most of the writing, with an occasional page of pithy practical comments by Callahan, an experienced conventional contractor. Snell's writing is intelligent but repetitious, particularly when he gets into philosophizing. The photographs are good, too, but many of them are used multiple times without good reason.

The heart of the book is the construction process, which is shown and explained in detail. The authors' techniques when building cob, cordwood, and stick frame walls are generally sound. And the lovely living roof is great to see demonstrated step-by-step. (Although I have read that fire ants can and do eat through ethylene propylene diene terpolymer (EPDM) roofing membranes, which are used in living roofs.)

However, I felt that the strawbale design, and the strawbale construction methods, were not based on a thorough understanding of the material. And this could have potentially serious consequences.

In a mixed-humid climate like that of North Carolina, bales are at their most vulnerable. It rains a lot, and if bales get wet, the humidity prevents them from drying very fast. Yet Snell and Callahan placed the strawbale wall on the gable end of the building, where the skimpy overhang scarcely protects it. And the construction also includes a stem wall just a few inches above grade, where rain is likely to pour off a nearby hill. Furthermore, because it is located on the north side of the house, the bale wall will never benefit from the drying heat of the sun.

Rightly concerned about splashback and ground moisture, Snell decided to completely wrap the first course of bales in Tyvek house wrap to keep out the rain. House wrap, a permeable fabric, is designed to repel liquid water but allow moisture vapor to move through. In practical applications, water vapor can sometimes condense inside the house wrap, where the moisture becomes trapped next to biodegradable materials.

Unfortunately, a sheet-metal termite barrier was installed under a nearby post. Rain and moisture collecting on this barrier will drain onto the foundation under the first course of bales. Also, during installation, the builders punctured holes in the moisture barrier that wraps the bales. In my opinion it is likely that a rainy spell in the not-too-distant future will deliver enough moisture to the bale wall that some will find its way inside the house wrap. There it will sit, unable to evaporate. And after a couple of weeks in weather above 40ºF, fungus will begin to grow, and the strawbale will be in trouble. I wouldn't imitate the rest of their strawbale wall-building technique, either, as it makes extra work out of a simple wall assembly.

For a big book, *Building Green* leaves a few things out. It inexplicably makes no mention of how much the building cost, or of how many hours it took to build it. Nor is there any mention of the building codes that govern these alternative techniques. Still, there is much of value and much inspiration in *Building Green*, and for only $30 it deserves a place on your bookshelf. But before you crack it open, I recommend a thorough reading of Bruce King's book.

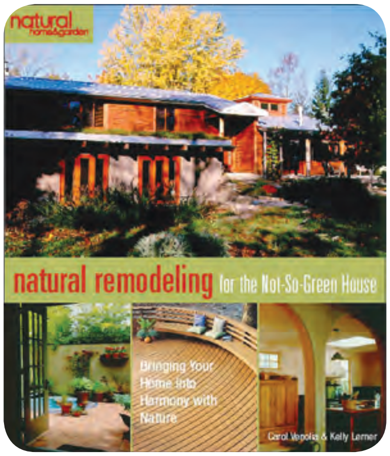
Snell, Clarke, and Tim Callahan. *Building Green: A Complete How-to Guide to Alternative Building Methods*. Asheville, North Carolina: Lark Books, 2005. 611 pp. paper. Illustrated. $30.

Natural Remodeling for the Not-So-Green House: Bringing Your Home into Harmony with Nature

It's nice to build your dream green home from the ground up, but often the most ecological thing to do is to remodel the house where you live now. It makes use of existing resources, reduces encroachment on nature, and usually saves money. *Natural Remodeling for the Not So Green House: Bringing Your Home into Harmony with Nature*, by architects Carol Venolia and Kelly Lerner, is a gold mine of ideas for affordably renewing your personal space.

Natural Remodeling tells how to increase the vitality of your home's biosphere, with life-enhancing effects for its human residents. The process begins with taking stock of your home as it is now, and where it is on the planet. To do this, you must first understand your climate, the challenges it poses, and its resources.

One of the many useful charts in this book divides North America into climate zones. The book also contains many useful lists, including a predesign questionnaire; comparison tables for selecting green materials;



and a regenerative checklist for design and construction. This allows you to develop your own natural strategies for increasing the comfort and energy efficiency of your home.

The book also contains instructive case studies, describing remodels in various different climate zones. These case studies illuminate the creative process by telling the people stories behind each remodel. Usually homeowners start out by believing that they need more space. Then they discover ways to make existing space better. A la *The Natural Step*, the authors suggest plucking the low-hanging fruit, with low-cost ideas for virtually immediate improvement, before tackling major projects. They also offer ideas on how to survive the upheaval that is a normal part of the remodeling process.

The book kept me turning pages with an attractive layout enhanced by over 400 color photos and numerous sidebars. Appendices, a glossary, and an extensive resource list support this useful reference book. The cumulative effect left me inspired and empowered. Even if you aren't planning to remodel, this book will make you more aware of ways to connect your home more fully to nature's vitality. And it expands the concept of a green home to a green lifestyle—one that brings greater personal enjoyment and provides an example for others to emulate.

Of course there is no one right way to build with strawbale or any other material. It only makes sense, before investing thousands of dollars in remodeling, or hundreds of thousands in building a home, to read and view a variety of resources. This investment in your education will save time and money during construction and will help you to fulfill your intentions in building a lasting, low-maintenance, and healthful home.

Venolia, Carol, and Kelly Lerner. *Natural Remodeling for the Not-So-Green House: Bringing Your Home into Harmony with Nature*. Asheville, North Carolina: Lark Books, 2006. 280 pp. paper. Illustrated. $25.

—Catherine Wanek

Author, photographer and filmmaker Catherine Wanek has been at the forefront of the strawbale building movement for more than a decade. Former publisher and editor of ***The Last Straw Journal****, she also produced the* ***Building with Straw*** *video series, and* ***The Straw Bale Solution*** *and* ***Urban Permaculture*** *videos. She is the author/photographer of* ***The New Strawbale Home****, a book of colorful photos and floor plans. Due out in Spring 2008, her next book will be* ***Hybrid Houses****.*

For more information:

To learn more about strawbale construction, go to www.strawbalecentral.com and www.thelaststraw.org.